Failure Analysis Report

FAIL-0829 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0829
Failure Type	Synchronization Failure
Severity	Critical
Component	Bluetooth Module
Application	APP-007 — InternetBrowser
Build Version	6.0.0
Timestamp	2023-10-02 23:33:11
Environment	ENV-003 — Hub Lab
OS Version	Tizen 7.0
Network	WiFi-6
Resolution	Open

2. Test Case Reference

Test Case ID	TC-1220
Title	Confirm that data synchronization triggers the expected notification when two co...
Execution ID	EXEC-04345
Test Run	RUN-011 — Regression Cycle 11 — Build 6.3.9
Requirement	REQ-0473 — UI Behavior — Diagnostics Report Generation

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete Bluetooth reconnection within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the server returns a 503 Service Unavailable respo...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-4458'; last_modified timestamps diverge by 7s.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.SensorManager.execute(line 759)
```

Log Reference	LOG-61760

5. Executed Steps at Time of Failure

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1	Enter valid credentials: username='testuser_039', password='[REDA...	A confirmation notification appears on the paired device.
2	Record response time using the built-in performance monitor.	The fallback UI is displayed with a human-readable error message.
3	Initiate NFC tap-to-pay by pressing the 'Pay' button.	The fallback UI is displayed with a human-readable error message.
4	Capture a screenshot and attach as evidence.	The fallback UI is displayed with a human-readable error message.

6. Root Cause Analysis

Race condition in the connection manager when two threads attempt simultaneous reconnection.

8. Historical Similar Failures

HIST-0702	FAIL-0703	Synchronization Failure	4.1.8	2021-12-29	7	Resolved
HIST-0828	FAIL-0829	Synchronization Failure	5.0.3	2021-04-21	4	Wont Fix
HIST-0830	FAIL-0831	Synchronization Failure	6.2.7	2023-06-24	2	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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